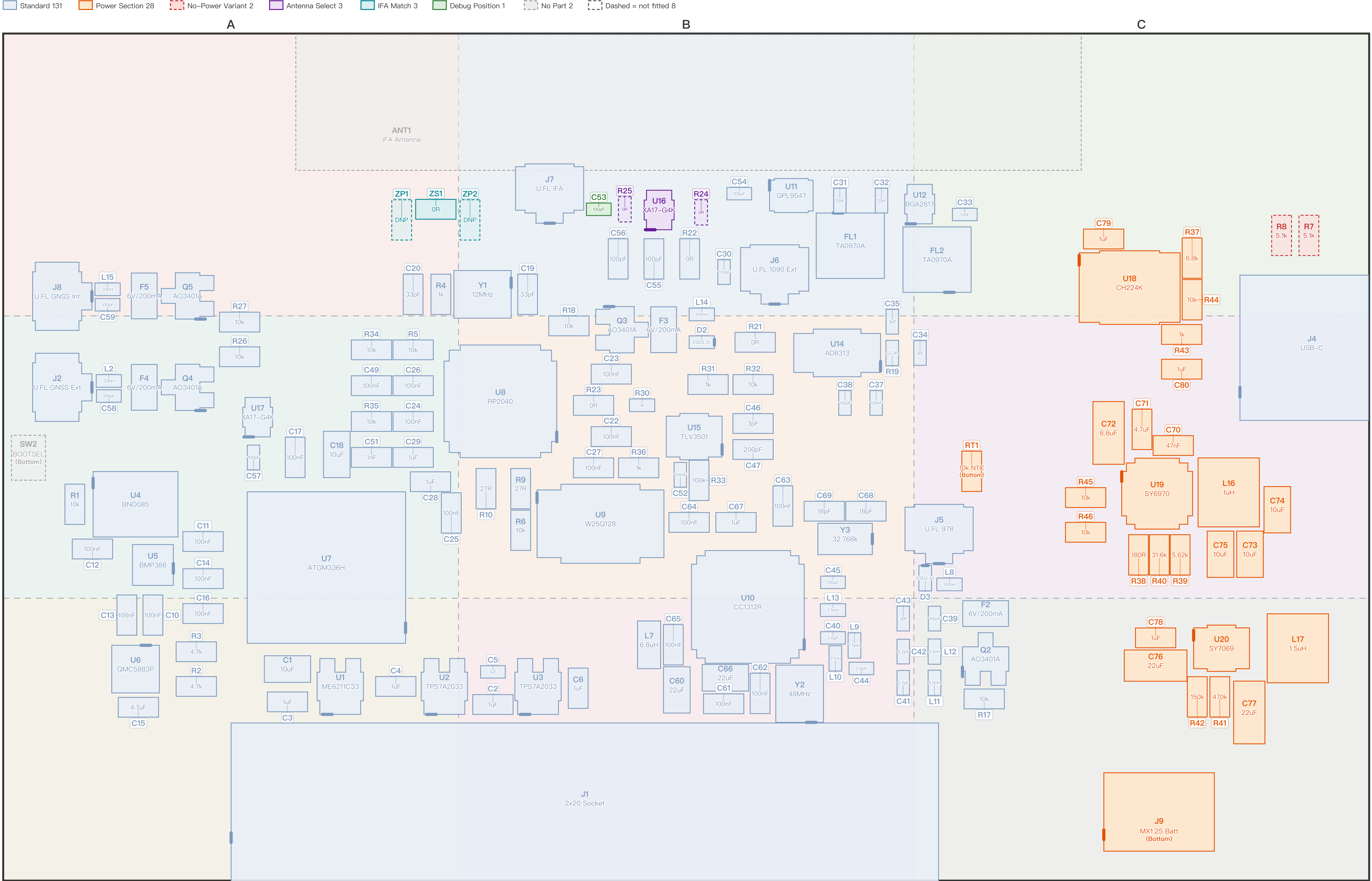


Pilot Kit Expansion Board V4.1 Assembly Map — Powered Variant (Top View)

Parts are drawn to their real outline and orientation; value inside the outline, designator outside. A heavy | marks pin 1 / the polarised end. The nine background tints (columns A/B/C, rows 1/2/3) match the grid badges on page 2. Board 100.0x62.0mm, scaled 5.62x.



Pilot Kit Expansion Board V4.1 Designator Index — Grid Reference

Each grid badge is the same colour as its cell on page 1 — match the colour to jump straight to it. Flags: X = not fitted, | = polarised (line it up with the polarity mark on page 1). Designators marked * have a specific note on page 3.

Ref	Grid	Type	Flags	Value	Footprint	Ref	Grid	Type	Flags	Value	Footprint	Ref	Grid	Type	Flags	Value	Footprint	Ref	Grid	Type	Flags	Value	Footprint
ANT1*	A1	No Part	X	IFA Antenna	PCB copper	C49	A2	Standard		100nF	0603	J6	B1	Standard		U.FL 1090 Ext	U.FL	R33	B2	Standard		100k	0603
C1	A3	Standard		10uF	0805	C51	A2	Standard		1nF	0603	J7	B1	Standard		U.FL IFA	U.FL	R34	A2	Standard		10k	0603
C2	B3	Standard		1uF	0603	C52	B2	Standard		100nF	0402	J8	A1	Standard		U.FL GNSS Int	U.FL	R35	A2	Standard		10k	0603
C3	A3	Standard		1uF	0603	C53*	B1	Debug Position		100pF	0402	J9*	C3	Power Section		MX1.25 Batt (Bottom)	MX1.25–2P	R36	B2	Standard		1k	0603
C4	A3	Standard		1uF	0603	C54	B1	Standard		100pF	0402	L2	A2	Standard		33nH	0402	R37	C1	Power Section		6.8k	0603
C5	B3	Standard		1uF	0402	C55	B1	Standard		100pF	0603	L7	B3	Standard		6.8uH	0805	R38	C2	Power Section		180R	0603
C6	B3	Standard		1uF	0603	C56	B1	Standard		100pF	0603	L8	C2	Standard		100nH	0402	R39	C2	Power Section		5.62k	0603
C10	A3	Standard		100nF	0603	C57	A2	Standard		100pF	0402	L9	B3	Standard		7.5nH	0402	R40	C2	Power Section		31.6k	0603
C11	A2	Standard		100nF	0603	C58	A2	Standard		100pF	0402	L10	B3	Standard		27nH	0402	R41	C3	Power Section		470k	0603
C12	A2	Standard		100nF	0603	C59	A1	Standard		100pF	0402	L11	C3	Standard		6.8nH	0402	R42	C3	Power Section		150k	0603
C13	A3	Standard		100nF	0603	C60	B3	Standard		22uF	0805	L12	C3	Standard		6.8nH	0402	R43	C2	Power Section		1k	0603
C14	A2	Standard		100nF	0603	C61	B3	Standard		100nF	0603	L13	B3	Standard		7.5nH	0402	R44	C1	Power Section		10k	0603
C15	A3	Standard		4.7uF	0603	C62	B3	Standard		100nF	0603	L14	B1	Standard		100nH	0402	R45	C2	Power Section		10k	0603
C16	A3	Standard		100nF	0603	C63	B2	Standard		100nF	0603	L15	A1	Standard		33nH	0402	R46	C2	Power Section		10k	0603
C17	A2	Standard		100nF	0603	C64	B2	Standard		100nF	0603	L16	C2	Power Section		1uH	SRN4018	RT1*	C2	Power Section		10k NTC (Bottom)	0603
C18	A2	Standard		10uF	0805	C65	B3	Standard		100nF	0603	L17	C3	Power Section		1.5uH	SRN4018	SW2*	A2	No Part	X	BOOTSEL (Bottom)	Solder jmp
C19	B1	Standard		33pF	0603	C66	B3	Standard		22uF	0805	Q2	C3	Standard		AO3401A	SOT–23	U1	A3	Standard		ME6211C33	SOT–23–5
C20	A1	Standard		33pF	0603	C67	B2	Standard		1uF	0603	Q3	B2	Standard		AO3401A	SOT–23	U2	A3	Standard		TPS7A2033	SOT–23–5
C22	B2	Standard		100nF	0603	C68	B2	Standard		18pF	0603	Q4	A2	Standard		AO3401A	SOT–23	U3	B3	Standard		TPS7A2033	SOT–23–5
C23	B2	Standard		100nF	0603	C69	B2	Standard		18pF	0603	Q5	A1	Standard		AO3401A	SOT–23	U4	A2	Standard		BNO085	LGA–28
C24	A2	Standard		100nF	0603	C70	C2	Power Section		47nF	0603	R1	A2	Standard		10k	0603	U5	A2	Standard		BMP388	LGA–10
C25	A2	Standard		100nF	0603	C71	C2	Power Section		4.7uF	0603	R2	A3	Standard		4.7k	0603	U6	A3	Standard		QMC5883P	LGA–16
C26	A2	Standard		100nF	0603	C72	C2	Power Section		6.8uF	1206	R3	A3	Standard		4.7k	0603	U7	A2	Standard		ATGM336H	LCC–18
C27	B2	Standard		100nF	0603	C73	C2	Power Section		10uF	0805	R4	A1	Standard		1k	0603	U8	B2	Standard		RP2040	QFN–56
C28	A2	Standard		1uF	0603	C74	C2	Power Section		10uF	0805	R5	A2	Standard		10k	0603	U9	B2	Standard		W25Q128	SOIC–8
C29	A2	Standard		1uF	0603	C75	C2	Power Section		10uF	0805	R6	B2	Standard		10k	0603	U10	B3	Standard		CC1312R	QFN–48
C30	B1	Standard		100pF	0402	C76	C3	Power Section		22uF	1206	R7*	C1	No–Power Variant	X	5.1k	0603	U11	B1	Standard		QPL9547	DFN–8
C31	B1	Standard		12pF	0402	C77	C3	Power Section		22uF	1206	R8*	C1	No–Power Variant	X	5.1k	0603	U12	C1	Standard		BGA2817	SOT–363
C32	B1	Standard		12pF	0402	C78	C3	Power Section		1uF	0603	R9	B2	Standard		27R	0603	U14	B2	Standard		AD8313	MSOP–8
C33	C1	Standard		12pF	0402	C79	C1	Power Section		1uF	0603	R10	B2	Standard		27R	0603	U15	B2	Standard		TLV3501	SOT–23–6
C34	C2	Standard		3pF	0402	C80	C2	Power Section		1uF	0603	R17	C3	Standard		10k	0603	U16*	B1	Antenna Select		XA17–G4K	SOT–363
C35	B2	Standard		3pF	0402	D2	B2	Standard		ESD3.3V	0402	R18	B2	Standard		10k	0603	U17*	A2	Standard		XA17–G4K	SOT–363
C37	B2	Standard		100nF	0402	D3	C2	Standard		ESD3.3V	0402	R19	B2	Standard		52.3R	0402	U18*	C1	Power Section		CH224K	ESSOP–10
C38	B2	Standard		100pF	0402	F2	C3	Standard		6V/200mA	0805	R21*	B2	Standard		0R	0603	U19	C2	Power Section		SY6970	QFN–24
C39	C3	Standard		100pF	0402	F3	B2	Standard		6V/200mA	0805	R22*	B1	Standard		0R	0603	U20	C3	Power Section		SY7069	TSOT–23–6
C40	B3	Standard		3.6pF	0402	F4	A2	Standard		6V/200mA	0805	R23*	B2	Standard		0R	0603	Y1	B1	Standard		12MHz	3225–4
C41	B3	Standard		2.7pF	0402	F5	A1	Standard		6V/200mA	0805	R24*	B1	Antenna Select	X	0R	0402	Y2	B3	Standard		48MHz	3225–4
C42	B3	Standard		6.2pF	0402	FL1	B1	Standard		TA0970A	SMD3838–6	R25*	B1	Antenna Select	X	0R	0402	Y3	B2	Standard		32.768k	3215–2
C43	B3	Standard		3pF	0402	FL2	C1	Standard		TA0970A	SMD3838–6	R26	A2	Standard		10k	0603	ZP1*	A1	IFA Match	X	DNP	0603
C44	B3	Standard		3.6pF	0402	J1	B3	Standard		2x20 Socket	2x20 P2.54	R27	A2	Standard		10k	0603	ZP2*	B1	IFA Match	X	DNP	0603
C45	B2	Standard		100pF	0402	J2	A2	Standard		U.FL GNSS Ext	U.FL	R30	B2	Standard		1k	0402	ZS1*	A1	IFA Match		0R	0603
C46	B2	Standard		3pF	0603	J4	C2	Standard		USB–C	USB–C 16P	R31	B2	Standard		1k	0603						
C47	B2	Standard		200pF	0603	J5	C2	Standard		U.FL 978	U.FL	R32	B2	Standard		10k	0603						

Pilot Kit Expansion Board V4.1 Selective Placement Notes

Standard parts are fitted unconditionally and are not listed here. Every class below carries a precondition; getting it wrong breaks the function or the board — full reasoning in SELECTIVE_PLACEMENT–zh_CN.md.

No–Power Variant (2 parts, default here: DNP)

5.1k CC pull–downs, fitted on the no–power variant only. Together with U18 (CH224K) they skew the CC resistance, so PD negotiation may request 9V/12V into a board designed for 5V. Both sit inside a NO–PWR VER silkscreen box.

- R7 5.1k [DNP] Not fitted on this (powered) variant
- R8 5.1k [DNP] Not fitted on this (powered) variant

Antenna Select (3 parts, default here: See notes)

Three mutually exclusive ways to route the 1090 antenna path — fit at most one. R24+R25 together short the external antenna to the on–board IFA; U16 plus either bypass shorts a switch port.

- R24 0R [DNP] 0R bypass → hard–wires the external antenna; not fitted here
- R25 0R [DNP] 0R bypass → hard–wires the on–board IFA; not fitted here
- U16 XA17–G4K [Fit] RF switch (default option) — this is the one to fit

IFA Match (3 parts, default here: See notes)

Pi matching network for the on–board IFA. The three positions combine freely into an L or pi network and are not exclusive. The default ZS1=0R is a straight–through link, not a tuned match — final values come from a VNA sweep with the enclosure fitted.

- ZP1 DNP [DNP] Shunt, antenna side; not fitted by default
- ZP2 DNP [DNP] Shunt, radio side; not fitted by default. First sweep starts at 3.3pF
- ZS1 0R [Fit] Series position, default 0R link; start the first sweep at 3.6nH

Debug Position (1 parts, default here: Fit)

Normally fitted; removed only for one specific measurement.

- C53 100pF [Fit] Remove before sweeping the IFA through J7, otherwise the VNA sees the switch and everything behind it instead of the antenna. Refit after

Power Section (28 parts, default here: Fit)

Fitted on the powered variant only. On the no–power variant this whole group is omitted and R7/R8 are fitted instead.

- J9 MX1.25 Batt [Fit] Battery connector, on the bottom side
- RT1 10k NTC [Fit] NTC for battery temperature sensing, on the bottom side
- U18 CH224K [Fit] CH224K USB–C PD sink controller; it provides the CC pull–down internally
- Others in this class: C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, L16, L17, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, U19, U20

No Part (2 parts, default here: —)

Bare copper or a solder–bridge jumper — there is no part to fit.

- ANT1 IFA Antenna [DNP] The on–board IFA — it is the PCB copper itself
- SW2 BOOTSEL [DNP] BOOTSEL solder bridge, short it with tweezers; on the bottom side

Standard (131 parts, default here: Fit)

This whole class is fitted unconditionally; only the few with an extra caveat are listed. The remaining 127 just get fitted.

- R21 0R [Fit] 0R link; open it to probe the AD8313 detector output on its own
- R22 0R [Fit] 0R link; open it to drive the 1090 switch A pin by hand
- R23 0R [Fit] 0R link; open it to drive the 1090 switch B pin by hand
- U17 XA17–G4K [Fit] GNSS–side RF switch. There is no bypass resistor on this side, so leaving it out means no signal path at all